Product Data Sheet

Product Datasheet

AO2 Oxygen Sensor

Document Purpose

The purpose of this document is to present the performance specification of the AO2 oxygen gas sensor.

This document should be used in conjunction with the Product Data Handbook (autooxops.pdf) and the Product Safety Datasheet (PSDS 4).

The data provided in this document are valid at 20°C, 50% RH and 1013 mBar for 3 months from the date of sensor manufacture.

Output signal can drift below the lower limit over time. For guidance on the safe use of the sensor, please refer to the Product Data Handbook (autooxops.pdf).

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Key Features & Benefits

- Molex Connector
- Linear Output from 0-100% Oxygen

Technical Specifications

MEASUREMENT

Operating Principle	Partial Pressure Electrochemical
Measurement Range	0-100% O_2
Output*	9 - 13 mV in Air
Response Time (T_{90})*	<5 s
Response Time ($T_{99.5}$) See Note 1*	<40 s
Baseline Offset*	<20 µV
Linearity	Linear 0-100% O_2

ELECTRICAL

On Board Temperature Compensation	<2% O_2 equivalent variation from 0°C to 40°C
External Load Resitor	10 kΩ Minimum
Connector	3 Pin Molex header (MOLEX 22-29-2031)
Recommended Mating Part	Molex 3-Way Housing (MOLEX 22-01-2035) Molex Crimp Terminals (MOLEX 08-45-0110)

MECHANICAL

Weight	40 g (nominal)
Housing Material	Red ABS
Orientation	Any

ENVIRONMENTAL

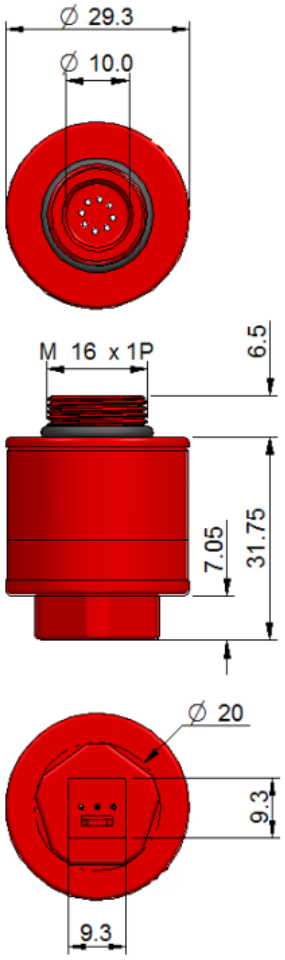
Typical Application	Vehicle Exhaust Analysis
Operating Temperature Range	-20°C to +50°C
Operating Pressure Range	0.5 - 2.0 Bar
Differential Pressure Range	0 to 500 mBar max
Operating Humidity Range	0 - 99% RH non-condensing

LIFETIME

Long Term Ouput Drift in 100% O_2	<10% signal loss/year
Expected Operating Life	360,000% O_2 hours at 20°C 286,000% O_2 hours at 40°C or 2 years in air at STP.
Packaging	Sealed blister

Note 1: $T_{99.5}$ Response is equivalent to a change in concentration fom 20.9% O_2 to 0.1% O_2

Product Dimensions



All dimensions in mm
All tolerances ±0.15 mm unless otherwise stated

IMPORTANT NOTE: Connection should be made via PCB sockets only. Soldering to the pins will seriously damage your sensor.

*** Specifications are valid at 20°C, 50% RH and 1013 mBar, using City Technology recommended circuitry. Performance characteristics outline the performance of sensors supplied within the first 3 months. Output signal can drift below the lower limit over time.**

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Poisoning

CiTiceLs are designed for operation in a wide range of environments and harsh conditions. However, it is important that exposure to high concentrations of solvent vapours is avoided, both during storage, fitting into instruments and operation.

When using sensors with printed circuit boards (PCBs), degreasing agents should be used before the sensor is fitted. Do not glue directly on or near the CiTiceL as the solvent may cause crazing of the plastic.

Mechanical Installation

When installing the sensor, it must only be screwed in hand-tight and a gas tight seal ensured. Spanners and similar mechanical aids may not be used, as excessive force may damage the sensor thread.

Cross Sensitivity Table

Whilst CiTiceLs are designed to be highly specific to the gas they are intended to measure, they will still respond to some degree to various other gases. The table below is not exclusive and other gases not included in the table may still cause a sensor to react.

IMPORTANT NOTE : The cross sensitivity data shown below does not form part of the product specification and is supplied for guidance only. Values quoted are based on tests conducted on a small number of sensors and any batch may show significant variation. For the most accurate measurements, an instrument should be calibrated using the gas under investigation.

The AO2 has been tested with a number of gases which may be present in automotive exhaust to establish their level of cross interference.

Gas	Concentration	Balance	$\%O_2$ Equivalent
Carbon Dioxide, CO_2	16%	N_2	<0.1
Carbon Monoxide, CO	6%	N_2	<0.1
Nitric Oxide, NO	3000 ppm	N_2	<0.1
n-Hexane	2000 ppm	N_2	<0.1
Hydrogen, H_2	5000 ppm	N_2	<0.1

SAFETY NOTE
This sensor is designed to be used in safety critical applications. To ensure that the sensor and/or instrument in which it is used, are operating properly, it is a requirement that the function of the device is confirmed by exposure to target gas (bump check) before each use of the sensor and/or instrument. Failure to carry out such tests may jeopardize the safety of people and property.

Every effort has been made to ensure the accuracy of this document at the time of printing. In accordance with the company's policy of continued product improvement City Technology Limited reserves the right to make product changes without notice. The products are always subject to a programme of improvement and testing which may result in some changes in the characteristics quoted. As the products may be used by the client in circumstances beyond the knowledge and control of City Technology Limited, we cannot give any warranty as to the relevance of these particulars to an application. It is the clients' responsibility to carry out the necessary tests to determine the usefulness of the products and to ensure their safety of operation in a particular application.